

TIM 172A, MOT-I, Lecture #2, 9/30/25

Agenda:

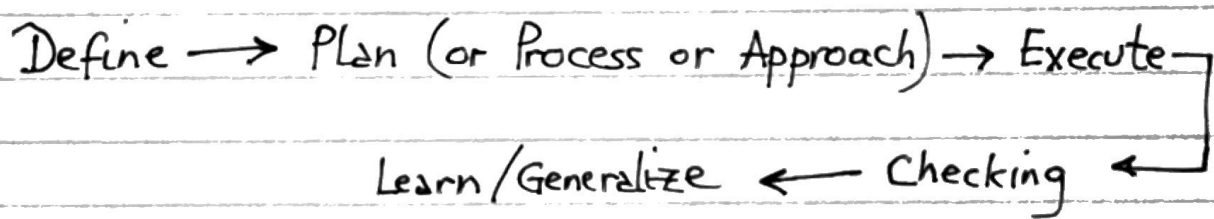
1. Quick summary of L#1
2. HW #1 (due on Thursday, 10/2/25)
3. TIM 172P
 - placed all students in group
 - posted the Project Proposal assignment
 - for the whole TIM 172A, the "FILES" tab on Canvas will contain a "TIM172P" folder

TIM 172A IMPORTANT NOTE:

- All submitted work must be TYPEWRITTEN
- We will NOT accept handwritten work
- Figures, graphics can be hand-drawn, & inserted in the typewritten text.

1. Quick summary of L#1

- Fast cycle competitors - FAANG & MANGA - can balance the conflicting trade-offs between product quality, speed, & cost; and can successfully manage risk.
- This course will develop the building blocks, & a framework - MDC - for integrating these blocks
- Professionals (Engineering, Business,) generally use a 5-step structured problem-solving process (SPSP):



2. HW # 1, Prob. # 1

Pre-liminary instructions (for all the work
- homeworks, exams, project tasks - done in
TIM 172A/P and beyond.) :

1. Develop a planning habit
 - Create a simple plan (what, when, how long, ...) for doing all the problems in an assignment
2. Develop a structured problem-solving habit.
 - Download the SPSP handout on Canvas
 - For every problem, "copy-and-paste" the handout into your proposed solution
 - Decide which sub-steps (for each of the 5 primary steps) are relevant or necessary to the solution of your particular problem
 - ⇒ Customize the SPSP for your particular problem.

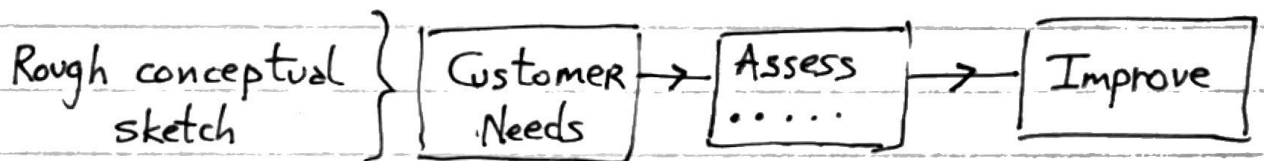
(back to) HW # 1, Prob # 1

Problem as given : "Improve the existing cell-phone."

⇒ Vague, open-ended statement

Apply the SPSP to this problem

Step 1 : Define the (real) problem
(requires reflection, thinking, asking questions)



Establish the sub-problems (SPs)

SP1 : Establish customer needs that all (generic) cell-phones should satisfy

SP2 : Assess existing cell-phone products to determine how well these customer needs are satisfied

w.r.t.:)
with
respect to

SP3 : Identify (generate) ways (or solutions) to improve existing cell-phones w.r.t.
(a) customer needs (from SP1) & assessment (from SP2)

Step 2 : Create a PLAN

(a) Assumptions

whom am I? : perhaps an engineer or
a marketing analyst at
a cell-phone company like
Apple, Samsung,...

who is the solution for : for the new product
development team, ...,
CTO, CEO

which type (manufacturer)
of cell-phones should
I (you) focus on } : OBVIOUS
("Be practical")

(b) What information do I need?

- how cell-phones work $\Rightarrow$ functions &
customer needs of
cell-phones, &
how they are
realized (made
real) or FORM

- other (?)

Create a plan for solving each sub-problem
(SP)

SP1 : Establish needs

Plan {
- how cell-phone work
- Internet research
- list of customer needs
 (functions, features, ...)
- ?
- ?

SP2 : Assess how well

{
- your own experience
- internet consumer research
- customer surveys, ...
- ...
- ...

SP3 : Improve

{
- research (internet)
- creative brainstorming
- ...?

Step 3 : Execute the plan

of the SPSP
process

SP1 : (from the plan steps) : execute : 1-2 hrs
plan

SP2 : execute : 2-4 hrs
plan

SP3 : execute : 2-4 hrs
plan

How should I present my results:
Table ?

examples

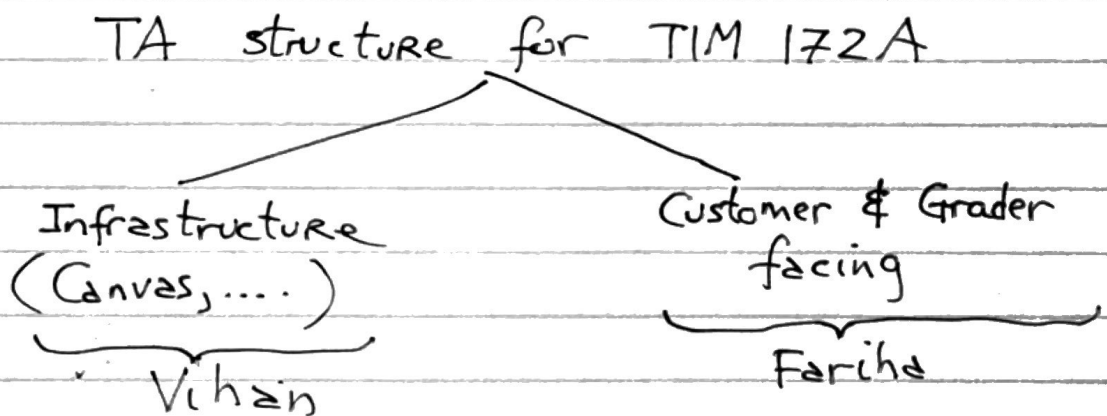
Needs	Assessment	?	?
ease of use			
performance			
reliability			
. . .			
. . .			

Step 4 : Check your work
(of the SPSP) (Reflect, ... ask questions, ...)

Step 5 : Learn & generalize
— about cell-phones & improving them
— yourself as a problem-solver

HW # 1 , Prob 2

simpler version of Prob 1



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